Picture Quiz: Identify the Constellations and Asterisms

Covers Lectures 2.3–2.5 about 60 minutes

Name: _____ Date: _____ Marks: _____ / 100

Allowed: nothing — Parts A and B are a memory test. For Part C you may use the hand-ruler values.

About the pictures

Every image was made with the planetarium program **Stellarium** from the real sky. Each shows its **field width** (bottom-left) and a small **N/E compass** (bottom-right). East is on the left when the image is the right way up. **In Part B many images are rotated**, just like the real sky at different times of night — use the compass and the shapes!

Word bank (some are unused, some used twice): Orion · Scorpius · Cassiopeia · Leo · Cygnus · Lyra · Gemini · Taurus · Canis Major · Boötes · Corona Borealis · Corvus · Crux · Great Square of Pegasus · Andromeda · Big Dipper (Saptarishi) · Little Dipper · Pleiades (Krittika) · Summer Triangle · Winter Hexagon · Teapot (Sagittarius) · Delphinus · Aquila · Perseus · Auriga · Hercules

Tips

Find the brightest star first — is it orange, red or blue-white? Look for three stars in a straight line (Orion's Belt) or a tight cluster (the Pleiades). Check the compass: the sky may be upside-down! Milky Way haze appears in summer and southern fields.

Part A — Stick figures without labels

12 images × 3 = 36 marks

For each image write (1) the **name**, (2) its **brightest star** (or any named star), (3) the **season** when it is best seen in the evening from India (winter / spring / summer–monsoon / autumn).



A1 Name: _____ Star: _____
Season: _____



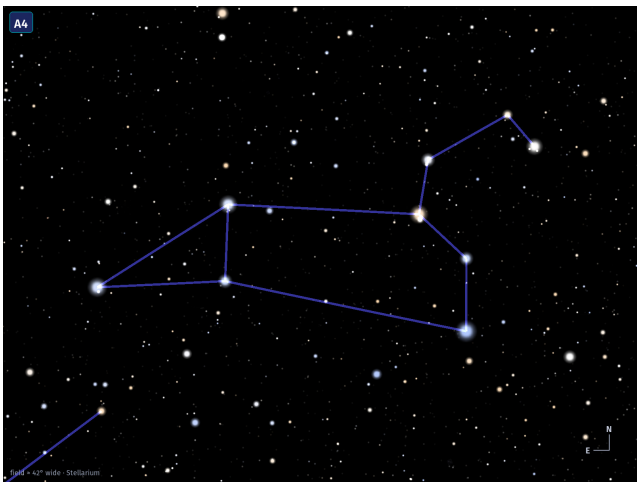
A2 Name: _____ Star: _____
Season: _____

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P.T.O.



A3 Name: _____ Star: _____
Season: _____



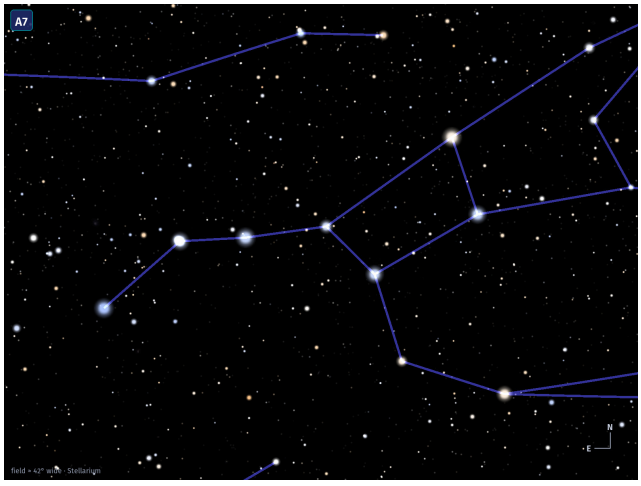
A4 Name: _____ Star: _____
Season: _____



A5 Name: _____ Star: _____
Season: _____



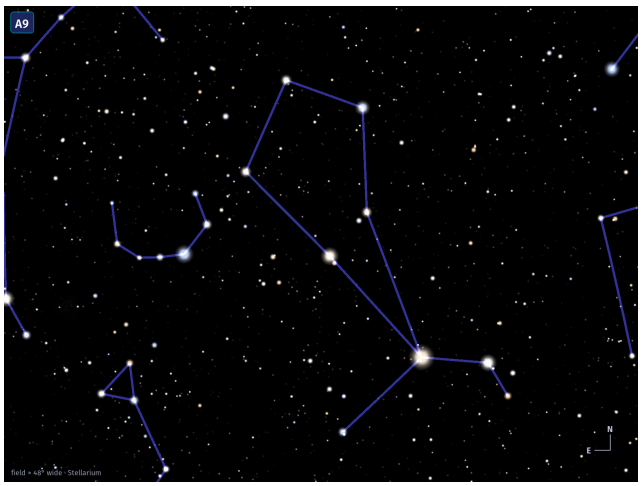
A6 Name: _____ Star: _____
Season: _____



A7 Name: _____ Star: _____
Season: _____



A8 Name: _____ Star: _____
Season: _____



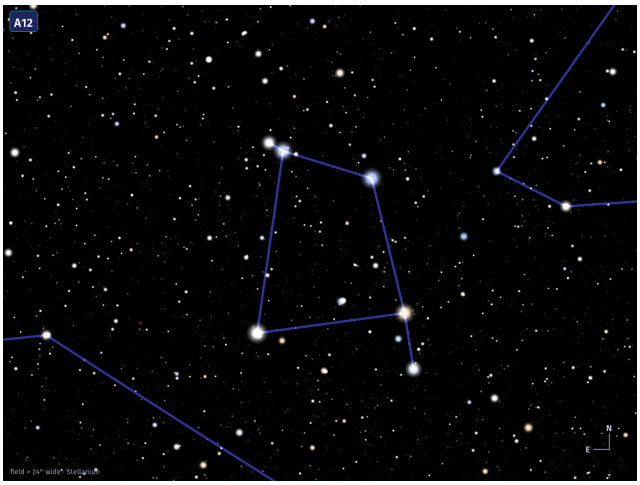
A9 Name: _____ Star: _____
Season: _____



A10 Name: _____ Star: _____
Season: _____



A11 Name: _____ Star: _____
Season: _____



A12 Name: _____ Star: _____
Season: _____

Part B — The real thing: no lines, often rotated ★★

14 images × 3 = 42 marks

For each image write (1) the **name**, (2) **C** if it is an official constellation, **A** if an asterism, or **Cl** if a star cluster, (3) **one clue** you used (a shape, a colour, a bright star…).



B1 Name: _____ C/A/Cl: ____ Clue: _____



B2 Name: _____ C/A/Cl: ____ Clue: _____



B3 Name: _____ C/A/Cl: ____ Clue: _____



B4 Name: _____ C/A/Cl: ____ Clue: _____



B5 Name: _____ C/A/Cl: ____ Clue: _____



B6 Name: _____ C/A/Cl: ____ Clue: _____



B7 Name: _____ C/A/Cl: ____ Clue: _____



B8 Name: _____ C/A/Cl: ____ Clue: _____



B9 Name: _____ C/A/Cl: ____ Clue: _____



B10 Name: _____ C/A/Cl: ____ Clue: _____



B11 Name: _____ C/A/Cl: ____ Clue: _____



B12 Name: _____ C/A/Cl: ____ Clue: _____



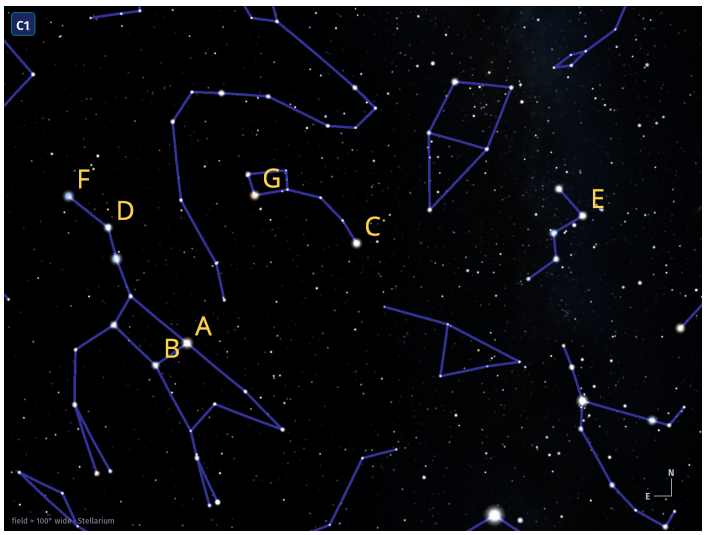
B13 Name: _____ C/A/Cl: ____ Clue: _____



B14 Name: _____ C/A/Cl: ____ Clue: _____

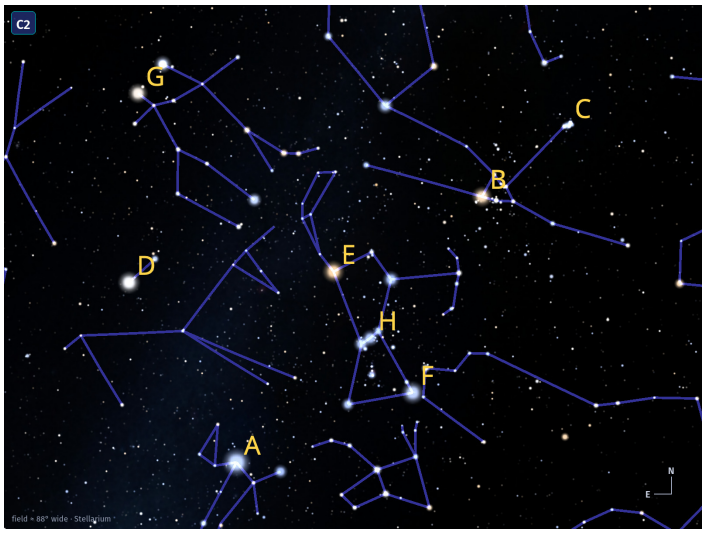
Part C — Star-hopping: name the lettered stars ★★ 22 marks

C1 — The northern sky (7 marks)



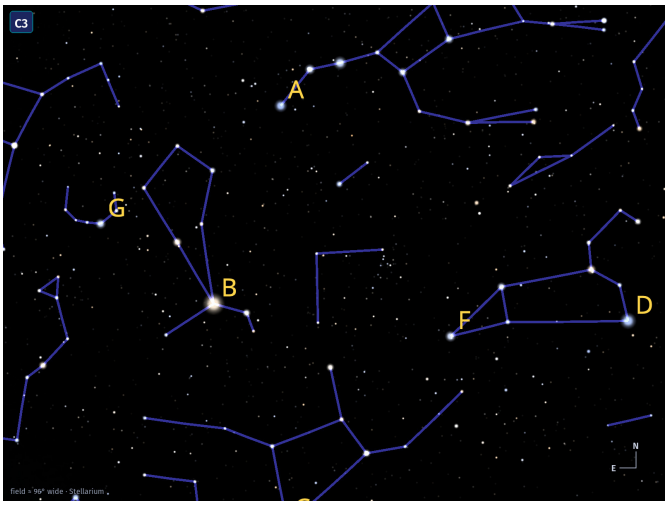
1. Name the stars: A _____ B _____ C _____ D _____ E _____ F _____ G _____ (½ each)
2. Which two lettered stars are the "pointers"? _____ and _____. Which star do they point to? _____ (1½)
3. Star D has a faint companion. What are their Indian names? _____ and _____ (1)
4. Are the Big Dipper and Cassiopeia circumpolar from Mumbai (19° N)? Explain using their declinations (≈ +49° to +64°). (1)

C2 — The winter sky (8 marks)



1. Name the stars/objects A–H. (½ each) A _____ B _____ C _____ D _____ E _____ F _____ G _____ H _____
2. Which three lettered stars form the **Winter Triangle**? (1½)
3. Which lettered star is **red-orange** and which **blue-white**, both in the same constellation? (1)
4. The line through H and its two neighbours points down-left to _____ and up-right to _____. (1½)

C3 — The spring sky (7 marks)



1. Name the stars: A _____ B _____ C _____ D _____ E _____ F _____ G _____ (½ each)
2. Complete: "Follow the **arc** to _____, then **speed on** to _____." (1)
3. Which lettered stars are the nakshatras **Swati**, **Chitra** and **Magha**? (1½)
4. G is the brightest star of a small semicircular constellation. Name it. (1)

Bonus ★★★ up to 6 marks

1. (2) Which Part B image shows a real physical group of stars born together, rather than a chance pattern?
2. (2) Images B1 and B7 both show “dippers”. Give two ways to tell the Big Dipper from the Little Dipper in a photo.
3. (2) Image B6 shows six bright stars in a ring with one more inside. Name the star inside and give its Indian name.